Zeyad Elshamy

AI Engineer | LLMs, AI Agents & Machine Learning

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SUMMARY

AI Engineer and Master's student with a strong foundation in machine learning and a passion for turning advanced AI concepts into **practical, intelligent systems**. Experienced in developing **LLM-powered agents, multi-agent architectures, and Retrieval-Augmented Generation (RAG) applications**, combining AI capabilities with robust Python backend development and API integration. Skilled in **LangGraph, LangChain, FastAPI, and modern AI frameworks**, with a focus on building reliable, scalable, and real-world AI solutions.

EDUCATION

Technische Hochschule Deggendorf (THD), Campus Cham 03/2025 – Present
M.Eng. Artificial Intelligence for Smart Sensors & Actuators *Cham, Germany*

- Relevant coursework: Machine Learning, Advanced Programming, System Design, Sensor Technology, and Edge-Device Architecture.

Arab Academy for Science, Technology & Maritime Transport 08/2020 – 09/2024
B.Sc. Artificial Intelligence (Robotics and Intelligent Systems) *Alamein, Egypt*

- GPA: 3.0/4.0; Dean's List. Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, and Reinforcement Learning.

PROFESSIONAL EXPERIENCE

Mercor – AI Training & Evaluation Contributor (Senior Writers) 10/2026 – Present
AI Model Training & Evaluation *Remote*

- Designed realistic, domain-specific tasks and structured workflows for **AI model training and evaluation**.
- Created reference deliverables and evaluation scenarios requiring technical reasoning and verifiable outputs.
- Reviewed task quality, consistency, and adherence to project guidelines and acceptance criteria.

TECHNICAL SKILLS

- **Programming & Data:** Python, SQL, C/C++, JavaScript, NumPy, Pandas , HTML , Css
- **AI Agents & LLMs:** LangGraph, LangChain, OpenAI API, Ollama, Llama 3.1, Tool Calling, Multi-Agent Systems, MCP, A2A, Prompt Engineering.
- **RAG & NLP:** Hugging Face Transformers, ChromaDB, Vector Databases, Embeddings, Semantic Search, Document Processing, Tokenization.
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, RNNs, Transformers, Model Evaluation, OpenCV.
- **Backend & Software Engineering:** FastAPI, REST APIs, Pydantic, JSON, OpenAPI/Swagger, API Integration, Git.
- **Cloud, DevOps & Tools:** Docker, Linux, GitHub Actions, AWS Fundamentals, Streamlit.

PROJECTS

VehicleConnect AI – Automotive Agent Integration Platform GitHub
Python, MCP, A2A, FastAPI, Docker, REST APIs, Pydantic

- Developed a containerized automotive integration platform connecting simulated vehicle telemetry and backend services through **MCP and REST APIs**.
- Implemented **MCP vehicle-data tools** and A2A communication with an independent charging-recommendation agent.
- Built an interactive web dashboard with vehicle monitoring, trip feasibility analysis, and charging recommendations.
- Designed authenticated REST endpoints with JSON validation and error handling, and integrated backend services using Docker.

Industrial Maintenance AI Agent GitHub
Python, LangGraph, LangChain, FastAPI, Tool Calling

- Developed an **LLM-powered industrial maintenance agent** using LangGraph and structured tool calling to retrieve machine status and analyze sensor readings.
- Built a **FastAPI backend** for natural-language interaction with simulated industrial equipment data, including temperature, vibration, and pressure.
- Implemented tools for machine-status retrieval, comparison against normal operating ranges, and maintenance recommendations.
- Generated structured fault explanations, abnormal-condition assessments, and suggested corrective actions based on tool responses.

Multi-Document RAG AI Assistant GitHub

Python, LangChain, Ollama, Llama 3.1, ChromaDB

- Developed a local **Retrieval-Augmented Generation (RAG)** assistant supporting question answering across PDF, DOCX, CSV, XLSX, PPTX, and TXT documents.
- Implemented document ingestion, text chunking, embedding generation, and **semantic retrieval** using ChromaDB and Nomic Embeddings.
- Integrated Llama 3.1 through Ollama for private, offline document question answering with source-grounded responses and citations.

RESEARCH & PUBLICATIONS

RAFEQI: Design and Implementation of a Cost-Effective Personal Assistant Robot 03/2025

RIMC, AASTMT Publishing Centre *Published Paper*

- Co-authored a peer-reviewed publication on a low-cost humanoid assistant robot integrating NLP, computer vision, sensor systems, and autonomous interaction.

LANGUAGES

- Arabic – Native | English – Fluent (C1) | German – A2 (Currently Improving).